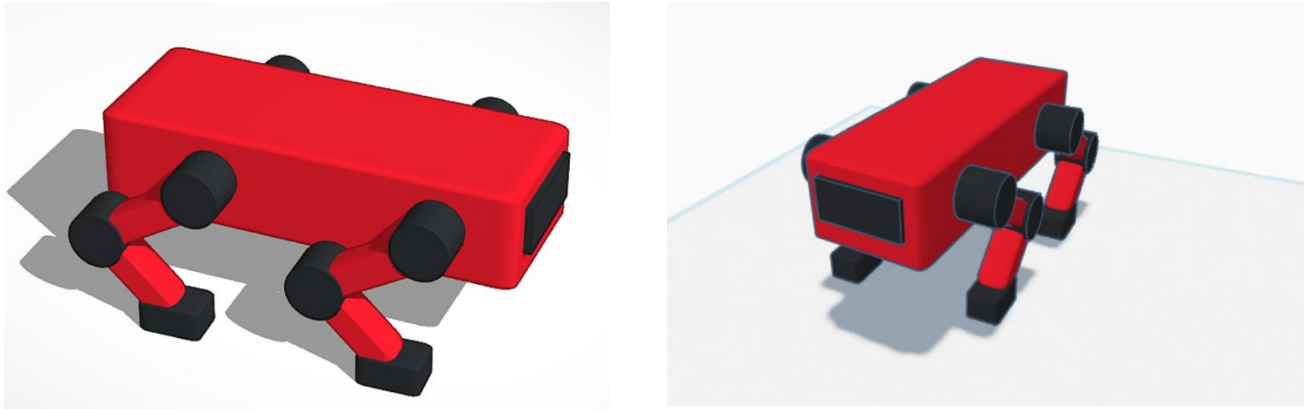


# Initial Mechanical Design of a Robotic Dog



*Figure 1: Initial 3D mechanical design of the robotic dog from different views.*

## 1. Body and Frame Design

I designed the initial robotic dog model using Tinkercad. The robot uses a simple rectangular rigid body as the main frame. The frame supports the four legs and joints and provides a stable structure for the robot. The rectangular shape was selected because it is simple to design and helps distribute the robot's weight between the four legs.

## 2. Leg Design

The robotic dog has four legs, and each leg consists of an upper leg, a lower leg, and a flat foot. The upper and lower sections are connected through a knee joint, while the leg is attached to the main frame through a hip joint. The angled leg design was selected to support the robot's weight and provide a suitable structure for standing and walking.

### 3. Number of Joints and Degrees of Freedom

Each leg has two joints: a hip joint and a knee joint. Each joint provides one degree of freedom (1 DOF), allowing rotational movement in one direction. Therefore, each leg has 2 DOF, and the four-legged robot has a total of 8 DOF.

### 4. Motor Selection

Servo motors are selected for the hip and knee joints of the robotic dog. A servo motor is used at each joint because the legs require controlled rotational movement. Since the robot has eight joints, eight servo motors are required, with one motor for each joint.

### 5. Initial Torque Calculation

An initial torque calculation was performed for the **hip joint**. Since the actual mass of the robotic dog has not been determined, the total robot mass is assumed to be **1 kg**. The weight is assumed to be equally distributed among the four legs; therefore, each leg supports approximately **0.25 kg**.

The gravitational force supported by one leg is calculated using:

$$\mathbf{F = mg} \quad \mathbf{F = 0.25 \times 9.81 = 2.45 \text{ N}}$$

The distance from the **hip joint, which is the axis of rotation, to the assumed point of force application** is taken as **0.03 m**.

The required torque is calculated using:

$$\mathbf{\tau = F \times r}$$

$$\mathbf{\tau = 2.45 \times 0.03 = 0.0735 \text{ N}\cdot\text{m} \approx 0.074 \text{ N}\cdot\text{m}}$$

Therefore, the minimum initial torque required at the hip joint is approximately **0.074 N·m**. This value represents the minimum torque under static conditions. In a real robotic dog, a motor with a higher torque should be selected to account for the weight of the leg itself, acceleration during movement, friction, and a safety factor.

## **6. Stability and Center of Gravity**

The robotic dog is designed with four legs distributed around the rectangular body to provide a wide support area and improve stability. The center of gravity is assumed to be located near the middle of the main body. Keeping the center of gravity within the support area formed by the four feet reduces the risk of the robot tipping over. The body is also kept relatively low to improve balance during standing and walking.

## **7. Proposed Walking Method**

A slow walking gait is proposed for the robotic dog. The legs move one at a time while the other three feet remain in contact with the ground to maintain stability. The hip and knee servo motors move each leg forward and then place the foot back on the ground. This sequence is repeated for the four legs to move the robot forward while keeping the center of gravity within the support area.

## **8. Expected Mechanical Challenges**

The robotic dog may face several mechanical challenges, including insufficient motor torque, high friction at the joints, poor alignment of the legs, and instability during walking. The frame or joints may also be affected by the robot's weight and repeated movement. These challenges should be considered when selecting the motors, designing the joints, and maintaining the center of gravity within the support area.

**Designed by: Aram Alhejaili**